

Eco-Link

A SOLAR-POWERED PISO WIFI SYSTEM
with Integrated Coin Capacity Alert

SRS

SOFTWARE REQUIREMENTS SPECIFICATION

ASTRYX



A. Software Requirements **Specifications**

Eco link: A Solar-powered Piso WiFi SSystem
with Integrated Coin Capacity Alert

Change History

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Introduction

This section describes the scope and overview of the Software Requirements Specification of Eco-Link: A Solar-Powered Piso WiFi System with Integrated Coin Capacity Alert. Also, this document includes discussion on the system overall description, IoT-based monitoring architectures, and specific technical requirements. Appendices and index are also included.

1.1. Purpose

This document aims to provide a detailed description of the requirements for Eco-Link: A Solar-Powered Piso WiFi System with Integrated Coin Capacity Alert to be implemented in community hubs, public plazas, and rural deployment areas.

This project is developed to replace the traditional, grid-dependent system of operating standalone Piso WiFi vending machines. The clients who compose of community members, students, commuters, and local residents, have no idea if a vending machine is operational or offline due to grid power outages, while the vending machine owners do not know when the coin boxes are full, how much power remains in the system battery, where a machine breakdown has occurred, and when do they need to collect the accumulated revenue. To get this data, they still need to conduct regular, physical site visits to inspect the operating condition of the aforementioned machines. Then the owners will have to manually count collected coins and troubleshoot power management issues before the actual internet vending service is finally restored or maintained.

Eco-Link is a solar-powered and IoT-integrated system that will allow clients to easily access internet connectivity by inserting coins into the physical hardware unit and authenticating through a localized captive portal page. Vending machine owners on the other hand will receive real-time system status updates and will be given remote access via mobile push notifications to monitor when a coin box reaches its full capacity threshold. A system administrator will manage the network bandwidth allocation, coin capacity alert parameters, solar power routing configurations, notifications, and real-time telemetry reports via a web-based dashboard system.

1.2 Scope

Eco-Link is a solar-powered, IoT-integrated hardware and network management system that will maintain the Vending Machine Node Status, Power Telemetry, Network Bandwidth Allocation, User Authentication, Coin Capacity Alerts, and System Reports. Specifically, the Eco-Link scope is outlined as follows:

Hardware & Vending Node Operations

- **Solar Power Harvesting & Power Resilience:** Utilizes a solar photovoltaic panel and battery storage coupled with a rule-based charge controller to ensure uninterrupted network uptime and automated source routing.
- **Real-Time Coin Capacity Monitoring:** Incorporates proximity or optical physical sensors

inside the vending enclosure to track the fill level of the coin box.

- **Edge Telemetry Node:** Captures operational metrics and transmits local system statuses to the cloud via cellular or backhaul network protocols.

Web & Mobile Application (Admin Dashboard)

- **Admin Privileges:**
 - Register and configure deployed Eco-Link physical hardware nodes.
 - Manage the following records and parameters:
 - **Vending Nodes** (Node ID, Deployment Location, Operational Status).
 - **Network Profiles** (Bandwidth Caps, Session Time Limits, Data Allocations per Coin Denomination).
 - **Coin Capacity Alerts** (Threshold Configuration, Calibration Curves, Recipient Contact Lists).
 - **Power Thresholds** (Low-Battery Disconnect Limits, Energy-Saving Deep Sleep Schedules).
 - **System Notifications** (SMS, Push Notification, or Email Templates).
 - Generate real-time and historical analytics reports by:
 - **Revenue Collection** (Daily, Weekly, Monthly earnings per node).
 - **Power Performance** (Battery State of Charge, Solar Generation vs. System Load Curves).
 - **Network Utilization** (Connected Users, Latency Metrics, Bandwidth Consumption Trends).
 - **Maintenance Logs** (Coin box full status occurrences, Hardware failure alerts).

User & Client Privileges

- **WiFi Customer Privileges:**
 - Connect to the local "Eco-Link" broadcasted SSID.
 - Access the localized pre-authentication Captive Portal landing page.
 - Insert physical coins into the vending slot to buy network credits.
 - Authenticate securely using alternative mechanisms such as temporary digital tokens or secure passkeys.
 - Receive real-time on-screen countdown timers indicating remaining connection time and bandwidth.
- **Vending Machine Owner Privileges:**
 - Granted permission to view individual node performance data.
 - Receive instantaneous mobile push notifications when a specific machine's coin capacity reaches its critical threshold limit.
 - Provided with exportable, printable revenue and telemetry summary sheets based on customized date criteria.

Term	Definition
Admin / Administrator	System user who is permitted to manage node parameters, configure configurations, monitor telemetry, and control the entire web-based ecosystem.
WiFi Customers	Users who are allowed to connect to the network, interact with the localized captive portal interface, and insert physical coins to purchase internet bandwidth and session time.
Vending Machine Owners	Remote users who are granted permission to access real-time node performance, view earnings telemetry, and receive automated coin capacity threshold notifications.
Web-based Application	A software package that can be accessed through a web browser. The software analytics engine and database reside on a central server and are accessed over a network.
Eco-Link	A collective term representing an ecologically sustainable network link; the name of the system which refers to a solar-powered, IoT-integrated Piso WiFi kiosk framework.
IoT (Internet of Things)	An interconnected system of physical edge microcontrollers and sensors that gather data from the physical vending environment and transmit it to a central web server.
Captive Portal	A localized web-authentication landing page that unauthenticated users are forced to interact with to validate credit tokens before obtaining wide-area internet access.
SQL	Structured Query Language. It is a standard computer language for relational database management, structured

	logging, and transactional data manipulation.
UML	Unified Modelling Language. It is a standardized modeling language enabling developers to specify, visualize, construct, and document artifacts of a hardware-software system.
SRS	Software Requirements Specification. It is a description of a system to be developed that lays out functional and non-functional requirements, and include a set of use cases that describe user interactions.
PHP / Node.js / Python	The primary server-side computing and backend scripting languages utilized to drive the edge routing telemetry, API handshakes, and dashboard views.
IEEE	Institute of Electrical and Electronics Engineers. The world's largest technical professional organization for the advancement of technology.

1.4 References

Books

[1]R. Stephens, *Beginning Software Engineering*, IN: John Wiley & Sons, Inc., 2015.

[2]F. Tsui and O. Karam, *Essentials of Software Engineering*, MA: Jones and Bartlett Publishers, 2011.

Websites

- IEEE Std 830-1998 (Revision of IEEE Std 830-1993)

Overall Descriptions

This section presents the overall description of Eco-Link in the context of how the administrator, vending machine owners, and WiFi customers interact with the system as well as how the system communicates with different hardware and software components. The system's essential features, functionality, constraints, assumptions, and dependencies will also be discussed.

Eco-Link is a solar-powered, IoT-integrated hardware and network management system that has the capability of maintaining the records of vending nodes, power telemetry data, bandwidth allocations, user authentication tokens, and remote coin capacity notifications.

The rest of the document provides an overview of the system functionality and system interaction with other network architectures. Likewise, it also introduces the different users and their interaction with the system as well as the system constraints and assumptions. The remaining chapter provides a detailed description of the hardware and software requirement specifications and various system user interfaces.

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2.1 Product perspective

2.1 Product Perspective - Eco-Link System Model

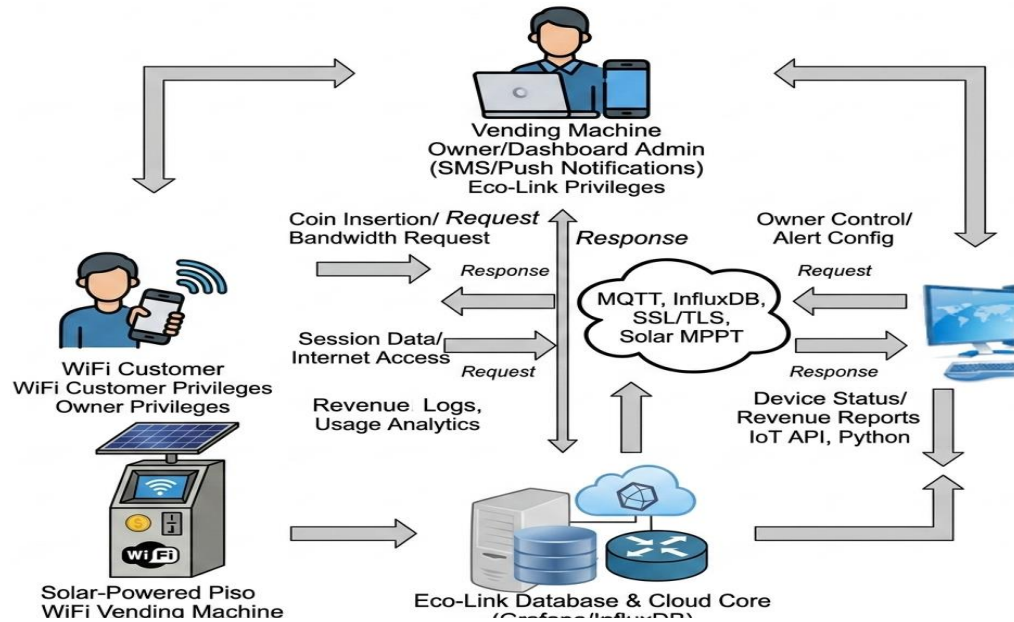


Figure 1 – Eco-Link Product Perspective and Data Flow Diagram**

As depicted in Figure 1, Eco-Link is a solar-powered, IoT-integrated hardware and network management system that will manage the internet connectivity access of WiFi customers as well as remote operational monitoring for vending machine owners. The Dashboard Administrator who is given overall system administration privileges can configure deployed physical hardware nodes, set data allocation policies per coin denomination, calibrate coin capacity thresholds, and manage system notifications. The WiFi customers are allowed to connect to the local wireless network, interact with a localized pre-authentication captive portal landing page, and insert physical coins to purchase high-speed internet session time. The vending machine owners can access their dedicated remote management page through pre-registered credentials to observe real-time system power diagnostics, track historical revenue collections, and instantaneously receive mobile push notifications the moment a specific machine's coin box hits its maximum storage capacity threshold.

2.2 Product functions

Eco-Link is designed to manage the remote operational needs of multiple vending machine systems as well as the training-related or status inquiries of the machine owners. It will likewise set an optimal operational framework and send real-time notifications to owners of the upcoming coin holder collection needs or system updates to be conducted. The system is purposely developed to answer the need of small-scale service technology operators for fast, accurate, reliable, and cost-effective data on coin capacity levels and efficient information dissemination on the active training or operational statuses available.

Unlike the present manual practice which is way too slow, tiring, and costly, **Eco-Link** is capable of managing, interpreting, and analyzing the coin capacity data and solar power telemetry digitally submitted by a large number of distributed vending nodes in seconds. Subsequently, the system is capable of generating valuable reports which can be used as inputs for community-responsive researches into off-grid resource sustainability. The same reports can be used as exhibits for accreditation of engineering or computing curricular programs.

With the aid of **Eco-Link**, the owners need not conduct house-to-house or physical machine surveys in every barangay to determine the physical capacity statuses of their systems. A printable report bearing the details of a particular vending station to be maintained and a master list of collected coin logs with their transaction information is provided for operators to utilize. The system provides a user-friendly interface for clients wherein they can simply access a captive portal, register their profile, and submit their transaction interest in the available WiFi programs.

Moreover, the system is capable of sending notifications to interested or affected owners only, even per client and node category. For quick reference, **Eco-Link** is embedded with a data visualization/dashboard feature which displays the number of active users, real-time power metrics, and coin capacity statuses in a particular service category in each barangay.

2.3 User characteristics

Three potential users are identified, classified, and described as follows:

System Administrator (Machine Owner / Operator)

The System Administrator is given full access to the following features:

- Add admin users, new vending node locations, payment thresholds, and user categories.
- Update system configurations, battery charge controller parameters, and error-handling thresholds.

- Set operational rules and maintenance schedules.
- Send notifications to machine owners or field technicians regarding coin capacity levels and system health.
- Print and generate telemetry reports per barangay, active node, and revenue/transaction category.

Node Maintainer / Field Technician (Owner)

The Owner or designated field technician is allowed to:

- View the data visualization dashboard showing the number of active network sessions, current battery voltage telemetry, and real-time coin capacity percentages.
- Receive mobile automated alerts when a coin box is full or when a diagnostic error code is transmitted.
- Confirm and log physical maintenance tasks, such as coin collection, hardware fault isolation, or relay replacements.
- Print printable reports containing individual machine status details, coin collection logs, and technical maintenance histories.

Clients (Community Members / WiFi End-Users)

The people in the community who use the services are allowed to take advantage of the following features:

- Access the local captive portal interface upon connecting to the broadcasted network.
- Register a user profile and manage secure network credentials.
- Search for available connectivity time slots or browse data package promotions.
- View descriptions of data limits, network speeds, and system operations.
- Convey transaction interest by inserting physical coins or interacting with hybrid payment interfaces.
- Receive real-time notifications regarding session balances, connectivity expirations, or system downtime warnings.
- Confirm network participation and initiate secure, authenticated internet access sessions.

User Characteristics and Assumptions

All users are assumed to possess the following:

- Ability to read and understand the English language or localized system prompts.
- Familiarity with the basic operations of standard smartphone interfaces (Android/iOS) or basic graphical user interfaces (GUI) for web browsing.
- Other than the above-mentioned characteristics, no further advanced facility with computer networking, solar engineering, or hardware technology can be assumed.
-

2.4 Operating Environment

The **Eco-Link** system will run natively in a hybrid environment consisting of a Real-Time Linux Operating System for internal mechatronic controls and cloud-based servers for data management. Since this is a web-based management and network-delivery system, the administrative dashboard and captive portal interface are expected to operate seamlessly across standard web browsers such as Google Chrome, Mozilla Firefox, and Safari on both desktop and mobile platforms.

Other than the central computing and local node hardware specifications—which include microcomputers like the Raspberry Pi or cost-effective ESP32/ESP8266 microcontrollers, dedicated relay modules, a hybrid coin/payment gateway, a localized OpenSSL-based validation engine, a centralized InfluxDB time-series data store, and a standard printer for administrative output logs—the hardware infrastructure relies heavily on a photovoltaic solar energy harvesting setup consisting of solar panels, batteries, and intelligent charge controllers to ensure complete off-grid autonomy. Furthermore, built-in Wireless Fidelity (Wi-Fi) routing modules and GSM capabilities are integrated into each vending unit to allow the system to broadcast network services to clients while transmitting distributed telemetry data simultaneously and seamlessly to the cloud database.

2.5 Assumptions and dependencies

The **Eco-Link** system will be used by the machine owners/operators, field maintenance technicians, and clients (community members and wireless end-users) from the different barangays for efficient, autonomous internet distribution, real-time coin monitoring, and remote system telemetry management. To maximize the utilization of the system, it is assumed that:

- The embedded source codes and algorithms governing power routing and

data logging are bug-free.

- The local captive portal and the administrator dashboard interfaces are user-friendly and responsive across devices.
- The users, node telemetry, and transaction logs are stored in a large, highly scalable time-series database server.
- A data visualization and search facility is provided on the dashboard for quick tracking of machine performance, error logs, and metrics.
- Administrators and maintenance technicians must have valid, secure credentials to gain access to the cloud backend, while clients must successfully pass through a localized captive portal authentication to access the network services.

System Dependencies

The system dependencies are listed as follows:

- The availability of minimum and specific hardware and software requirements, including local microcontrollers, operational solar harvesting components, and stable cloud network hosting.
- The unreliability of solar energy due to local weather conditions and environmental disruptions, which dictates the system's baseline power resilience.
- The data integrity and synchronization accuracy of real-time coin logs and telemetry status updates compiled by the distributed nodes into the centralized database.
- The compliance of data handling procedures with established legal regulations, specifically the Data Privacy Act of 2012, to maintain enterprise network security.
- The general understanding and basic digital familiarity of the community clients in manipulating smartphone settings to navigate and interact with a passwordless captive portal interface.

Specific Requirements

This section highlights the detailed discussion of the expected outputs given the required inputs for the **Eco-Link** system. Moreover, it gives a description of the hardware, software, and communication interfaces as well as the basic prototypes of the user interface designed to enable sustainable, autonomous, and remotely manageable internet vending operations.

3.1 External interface requirements

1. User Interfaces

Eco-Link provides a user-friendly graphical interface for ease of creating and updating operational parameters, tracking coin collection events, monitoring solar battery telemetry, broadcasting automated alerts, and print-previewing administrative performance logs.

- **Administrative and Owner Dashboard:** A web-based graphical interface (utilizing visualization tools like Grafana) that displays map distributions, time-series data plots for electrical telemetry, and system-wide notifications.
- **End-User Captive Portal:** A responsive, localized webpage that triggers automatically upon a client's connection to the Wi-Fi network, featuring passwordless data package registration and status tracking.

2. Hardware Interfaces

The specific hardware infrastructure is designed to bridge local mechatronic controls with renewable energy systems and remote data logging. The minimum hardware requirements are as follows:

- **Processor / Compute Nodes:** * Central Cloud Server: 2.3GHz multi-core processor (or cloud equivalent) for data aggregation.
 - Local Machine Controllers: Raspberry Pi microcomputer and/or ESP32/ESP8266 microcontrollers.
- **Memory (RAM):** 4GB (Server side) / 512MB to 1GB (Local microcomputer edge node).
- **Storage (HDD/Data Store):** 500GB SSD/HDD (Cloud server database storage).
- **Input/Output (I/O) Devices:** * **Inputs:** Multi-bay physical coin acceptors/hybrid payment gateways, solar panel voltage sensors, and temperature/capacity sensors.
 - **Outputs:** Automation relay modules (for load switching and fault isolation), physical Wi-Fi antennas, local status LEDs, and an

administrative office printer for physical report generation.

3. Software Interfaces

The **Eco-Link** system requires the following Software Specifications to manage autonomous processing and high data integrity:

- **Operating System:** Real-Time Linux Operating System (for edge vending nodes and robotics execution) and Linux/Windows Server environments (for cloud architecture).
- **Programming Languages & APIs:** Python/C++ (for embedded microcontrollers and mechatronic logic), JavaScript/HTML5 (for dashboard and captive portal frontends), and the WebAuthn API (for cryptographic passwordless authentication).
- **Database:** InfluxDB (Time-series database for real-time telemetry logs) or scalable distributed data stores (e.g., Apache Cassandra / Amazon DynamoDB configurations).
- **Database / Server Architecture:** Cloud-native clusters operating on deterministic, quorum-based synchronization protocols ($R + W > N$) to ensure causal or sequential consistency across multiple node replicas.
- **Security Middleware:** An OpenSSL-based validation engine paired with enterprise authentication logic (such as RADIUS or LDAP) for network access protection and certificate chain validation.

4. Communications Interfaces

Since **Eco-Link** is a web-based, distributed network delivery system, it requires robust communication pathways to seamlessly bridge remote local hardware with centralized management platforms.

Specifically, each vending node necessitates a built-in **Wireless Fidelity (Wi-Fi) module** to simultaneously broadcast local hotspot connectivity to community clients and establish a wireless interface for system transactions. To achieve continuous telemetry delivery and remote mobile notifications when nodes are deployed in off-grid or remote areas, the communication interface integrates a **GSM/Cellular modem**. This dual wireless approach ensures that data packets—such as real-time voltage, error codes, and coin capacity metrics—are seamlessly transmitted from the distributed local nodes to the cloud database server via TCP/IP protocols.

3.2 Functional requirements

3.2.1 Eco-link System admin and client interface use cases

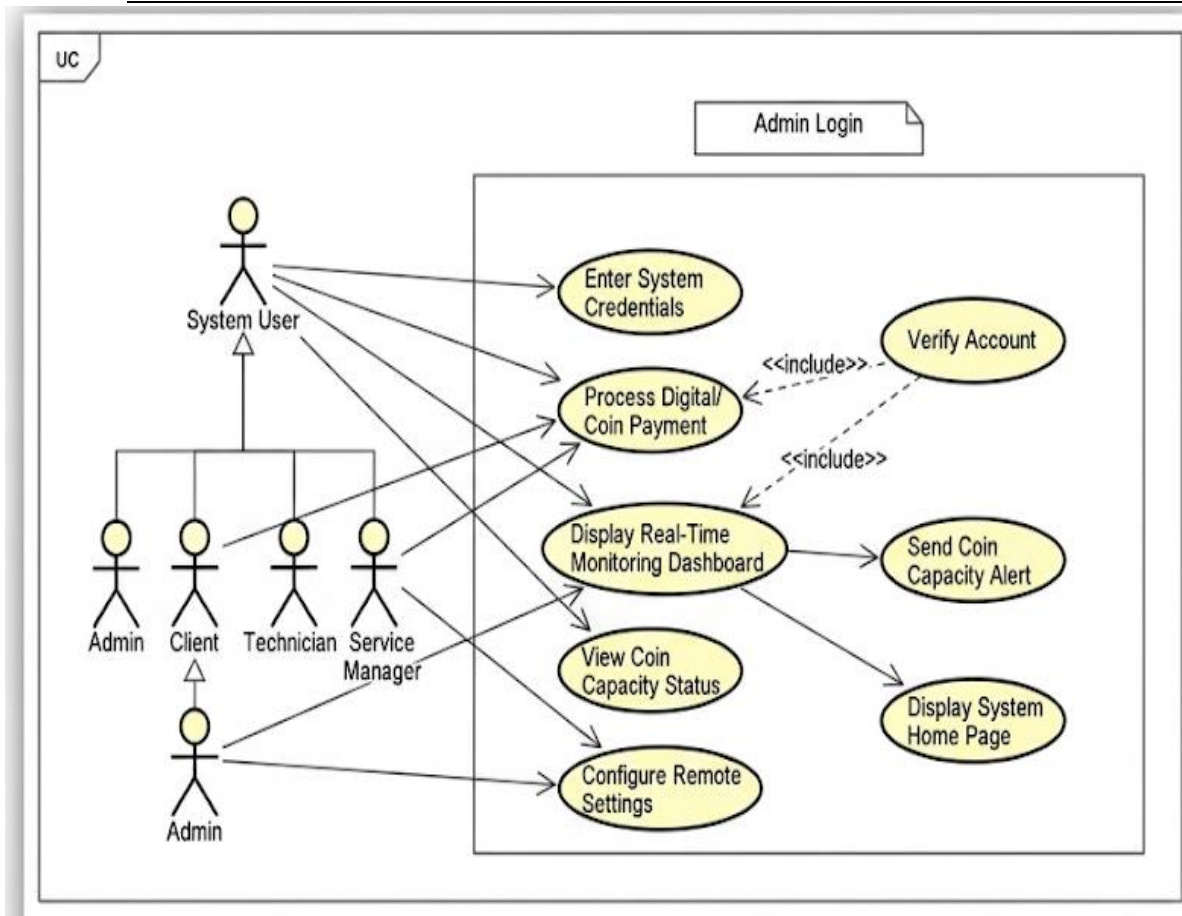
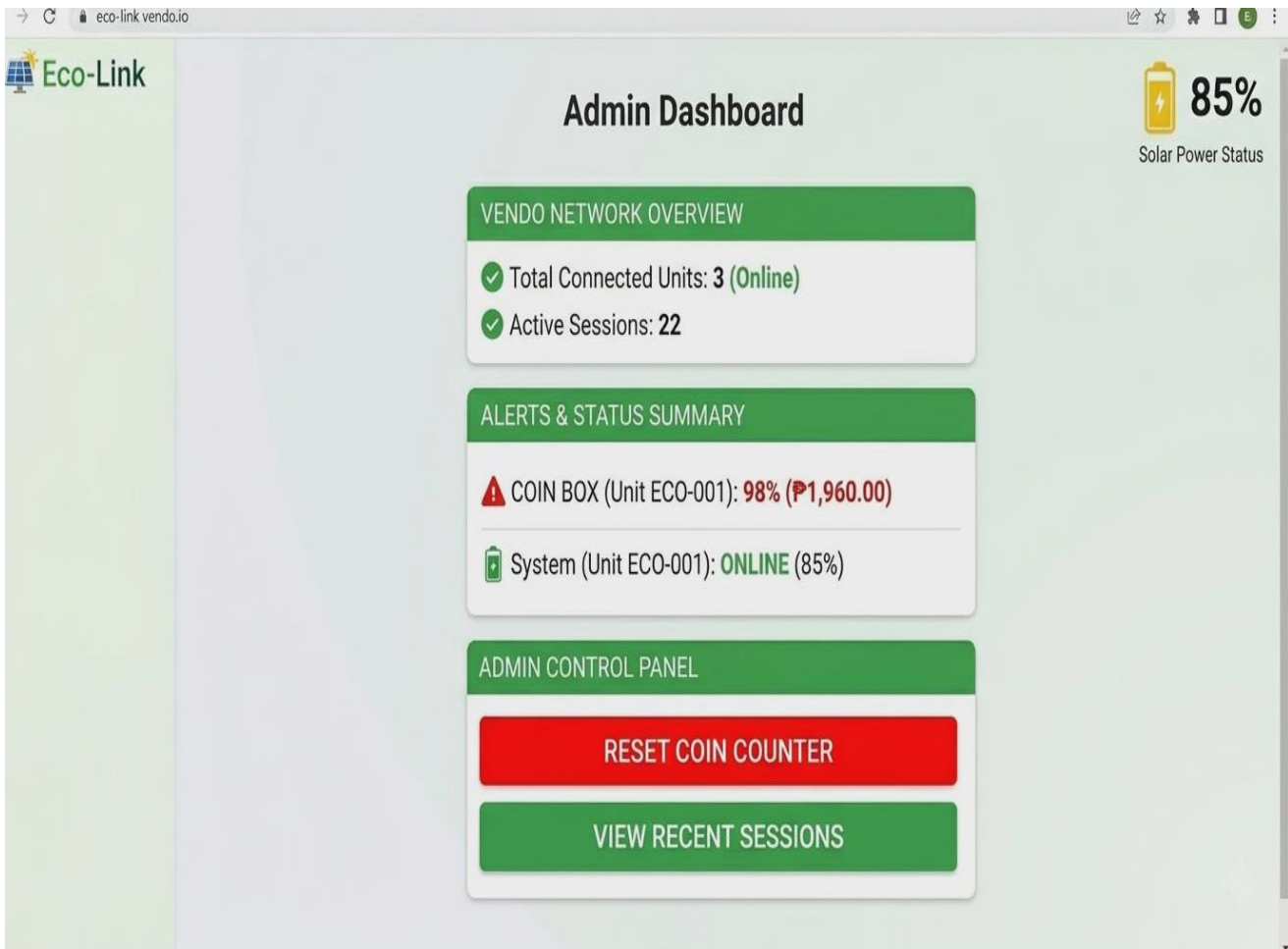


Figure 2. Eco-link system functionality use case diagram

3.2.2 Admin Login Interface Prototype

Figure 3. Admin Login dashboard interface



3.2.3 Admin Home Page Use case

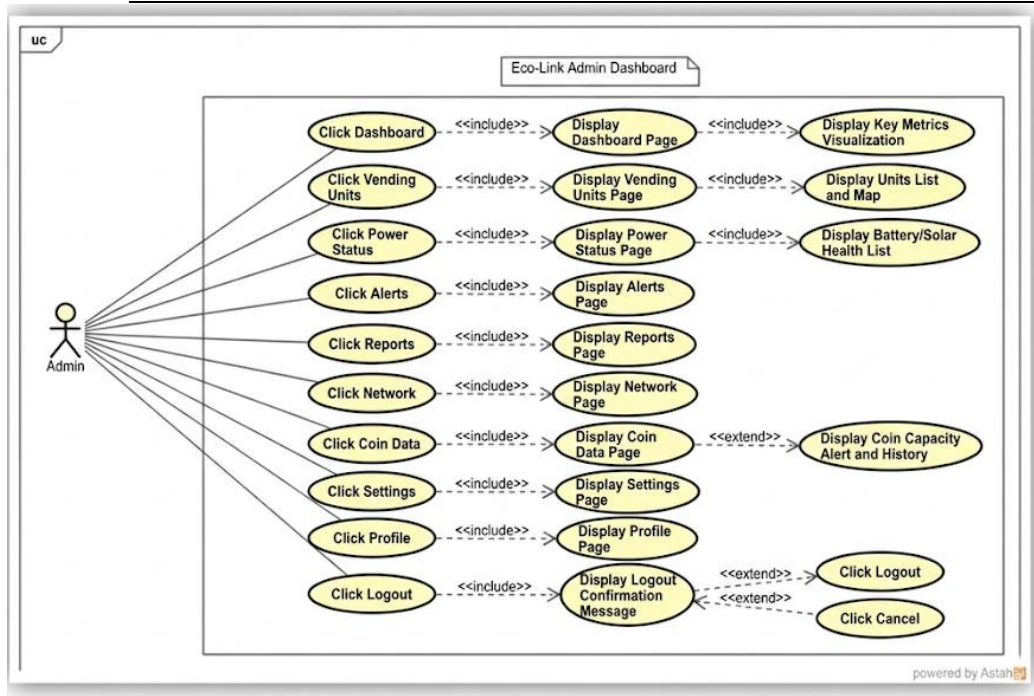


Figure 4. Admin dashboard use cases

3.2.4 Admin Home Page Prototype

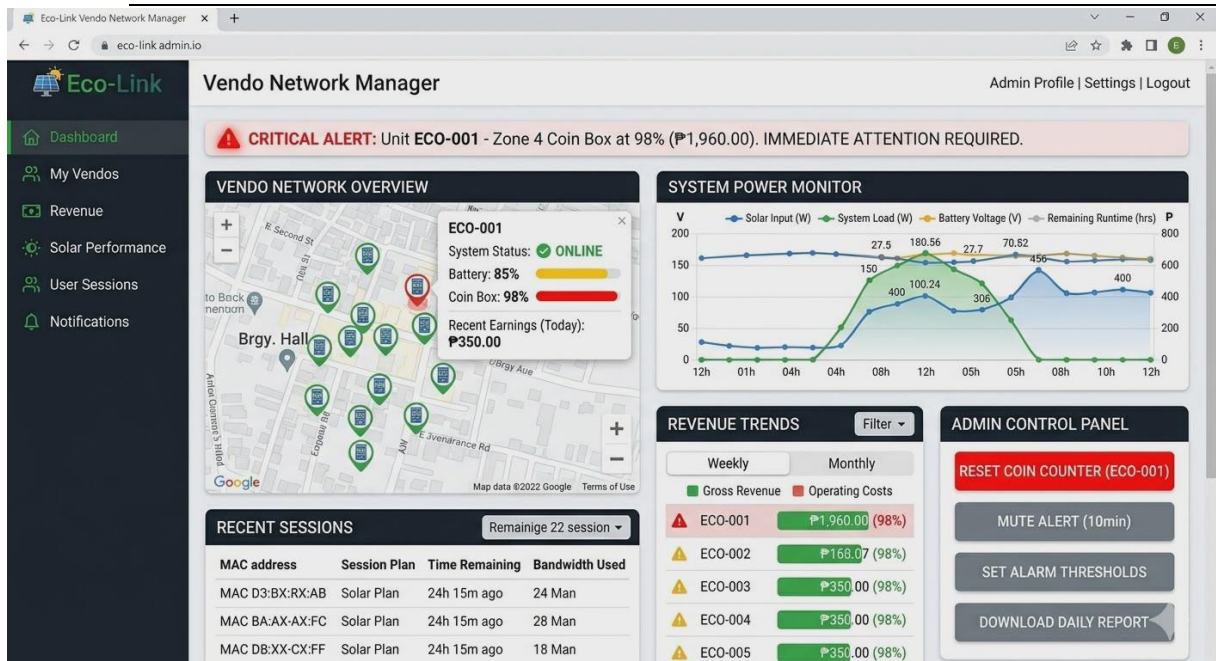
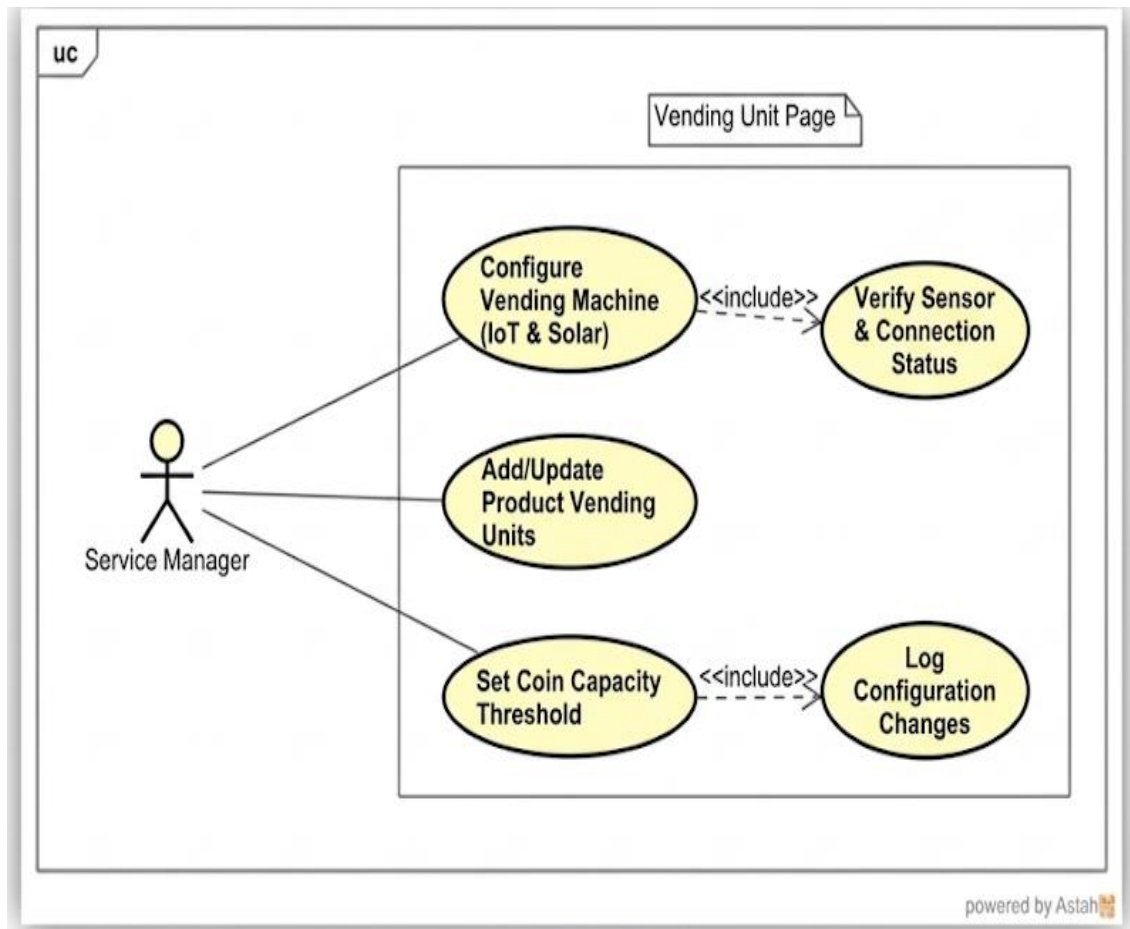


Figure 5. Admin mainPage

3.2.5 Manage vending units Use



case

Figure 6. Manage vending units Use Case

3.2.6 Manage Clients Prototype

Train Me

Clients Trainers Training Needs Notifications Reports Settings Profile Logout

CLIENTS

Show 5 ▼ records Search

ID Number	Name	Address	Contact Number	Category
1	Roger C. Aguilar	Abaca , S. E.	09195578911	Farmer
2	Dahlia E. Lamera	Cubay , S. E.	09275678901	PWD
3	Vilma C. Garcia	Dumiles , S. E.	09052678910	Housewife
4	Ely D. Garcia	Dumiles , S. E.	09167678791	Farmer
5	Edmund E. Baco	Garita, S. E.	09185699918	OSY

Showing 5 of 200 records

Previous 1 2 Next

Figure 7. Manage Client Prototype

3.2.7 Manage Trainers Use case

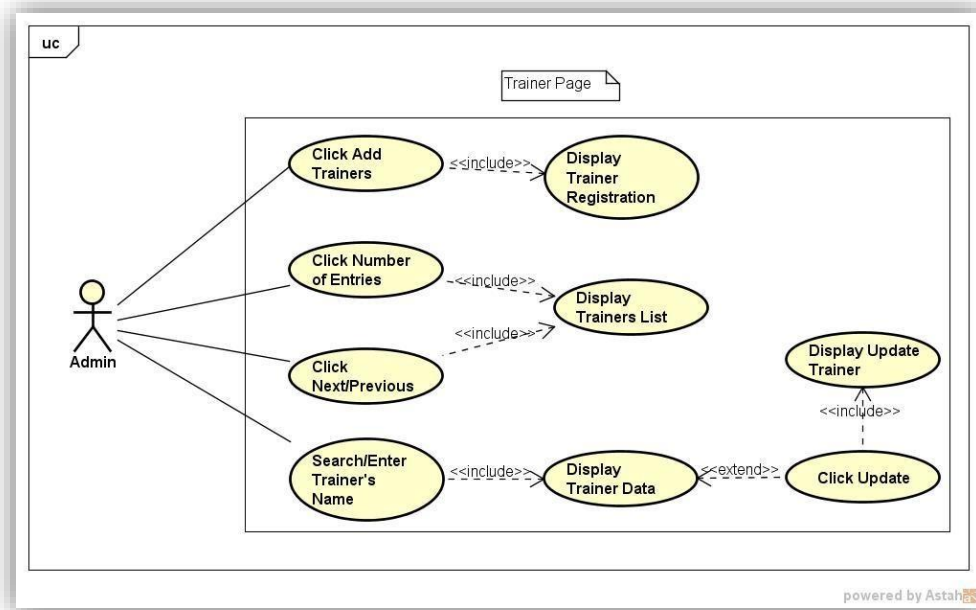


Figure 8. Manage Trainers Use Case

3.2.8 Manage Trainers Prototype

The prototype shows a web interface for managing trainers. It features a top navigation bar with links: Clients, Trainers, Training Needs, Notifications, Reports, Settings, Profile, and Logout. Below this is a section titled 'TRAINERS' with a '+ ADD TRAINERS' button. A 'show 5 records' label is present, along with a 'Search' input field. A table displays the following data:

ID NUMBER	NAME	SPECIALIZATION	ACTION
1	Weena J. Bulnes	IT	[Edit]
2	Katherine P. Padilla	Agriculture	[Edit]
3	Nicanor P. Parreño	Education	[Edit]
4	Pia C. Paz	Livelihood	[Edit]
5	Dyline P. Dacudao	I.T.	[Edit]

At the bottom, it says 'showing 5 of 40 records' and includes pagination controls: 'Previous', '1', '2', and 'Next'.

Figure 9. Manage Trainers Prototype

3.2.9 Trainer Registration Use case

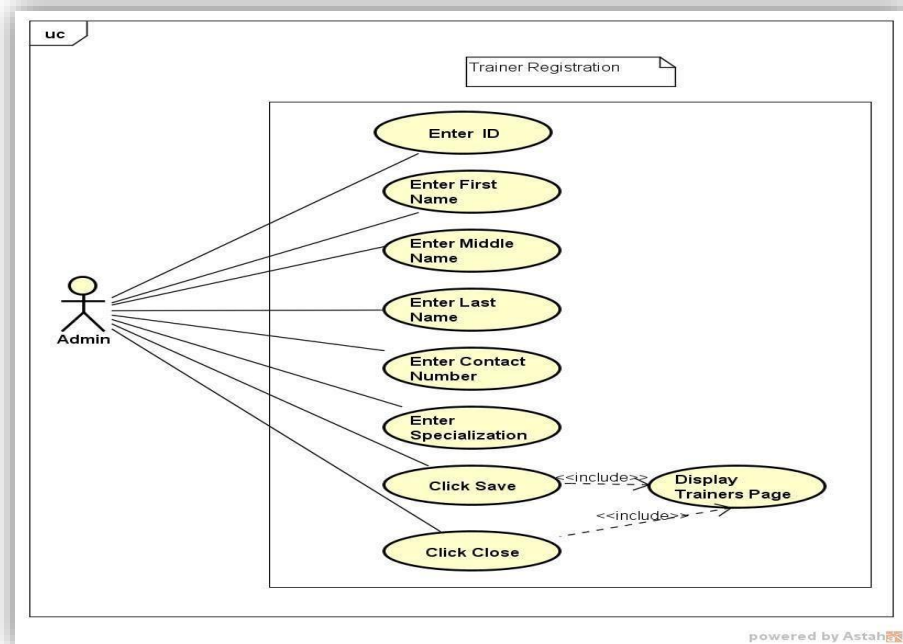


Figure 10. Trainer Registration Use Case

3.2.10 Trainer Registration Prototype

A hand-drawn prototype of a web form titled 'ADD TRAINERS' in a teal header box. The form is on a light blue background. It contains the following fields and controls:

- 'ID Number' with a text input field.
- 'First Name' with a text input field.
- 'Middle Name' with a text input field.
- 'Last Name' with a text input field.
- 'Specialization' with a dropdown menu (indicated by a small downward arrow icon).
- At the bottom right, there are two buttons: a teal 'Save' button and a white 'xClose' button with a black border.

Figure 11. Trainer Registration Prototype

3.2.11

Manage Training Needs Use case

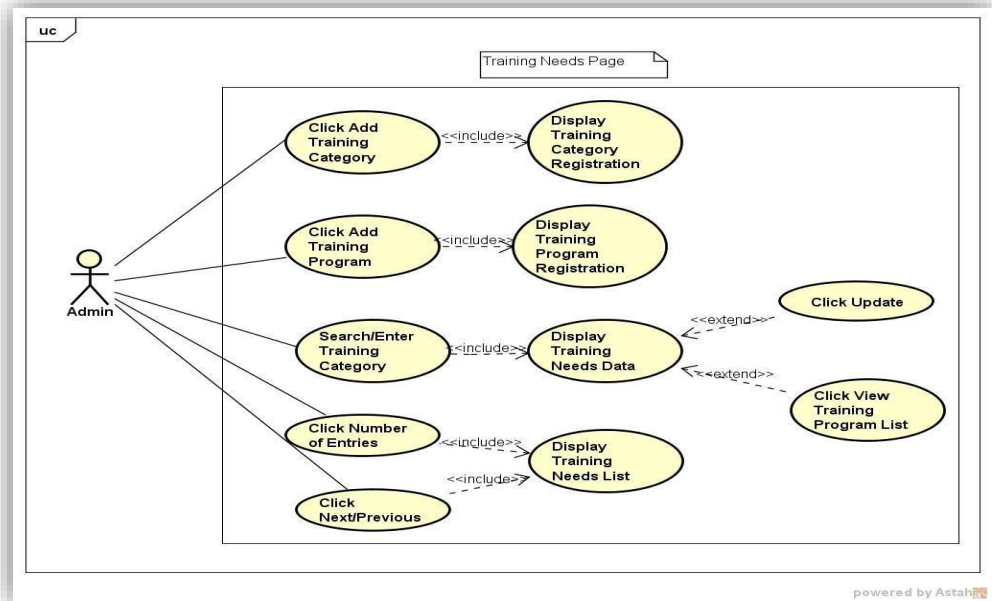


Figure 12. Manage Training Needs Use Case

3.2.12 Manage Training Needs Prototype

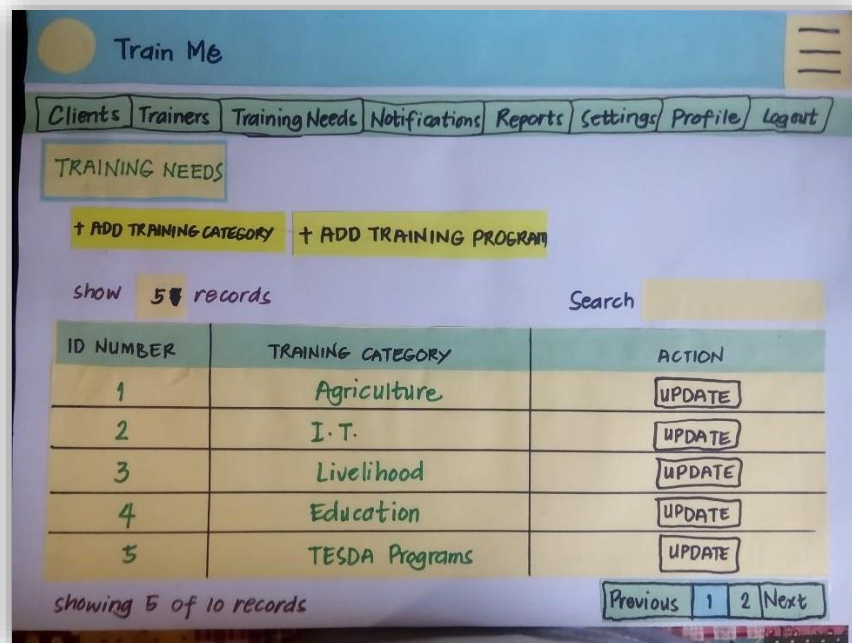


Figure 13. Manage Training Needs Prototype

3.2.13 Training Category Registration Use case

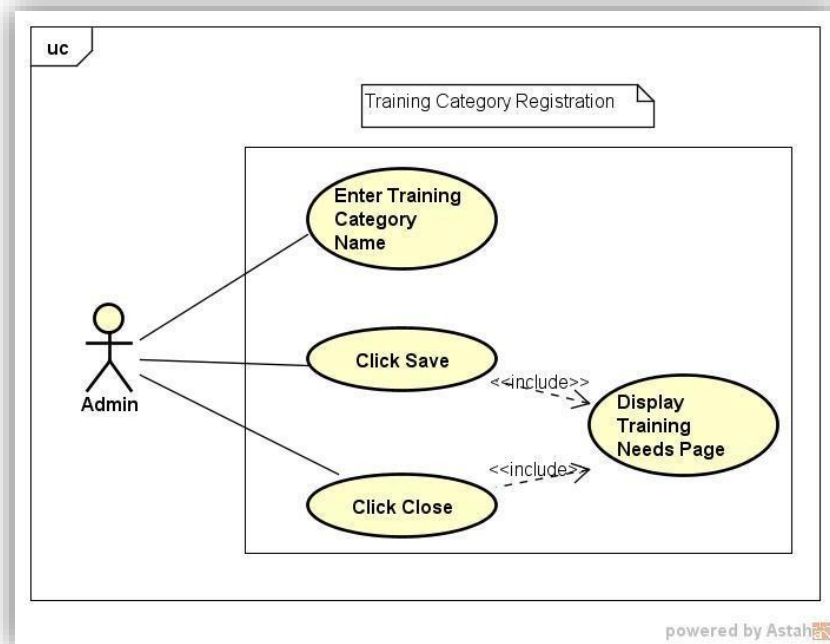


Figure 14. Training Category Registration Use Case

3.2.14 Training Category Registration Prototype

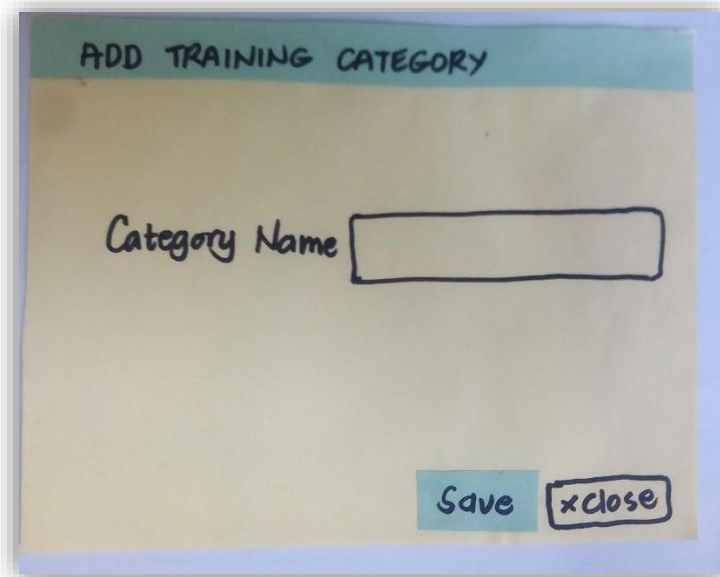


Figure 15. Training Category Registration Prototype

3.2.15 Training Program Registration Use case

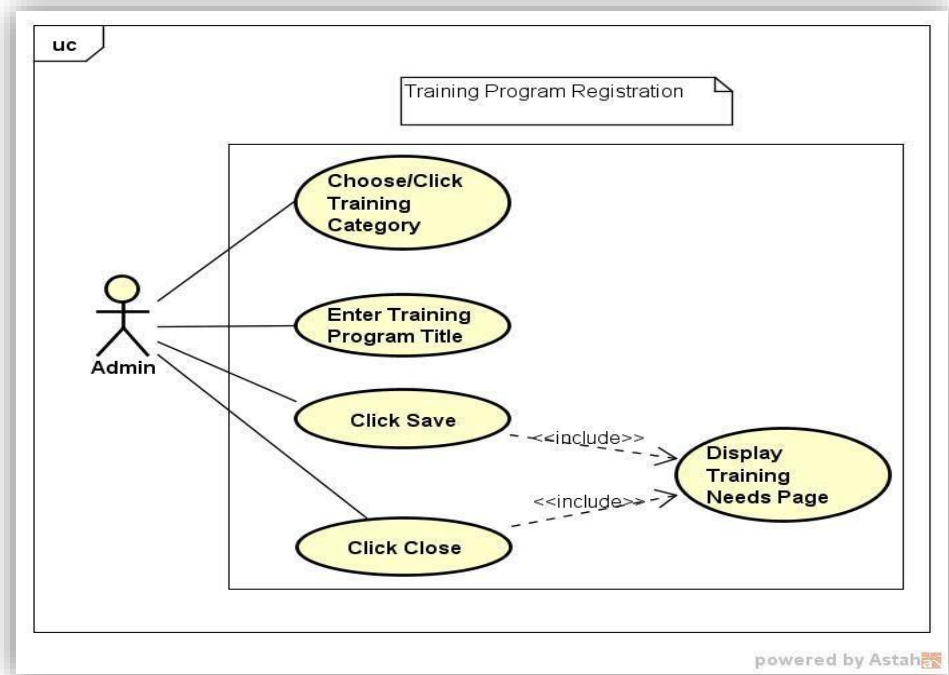


Figure 16. Training Program Registration Use Case

3.2.16 Training Program Registration Prototype

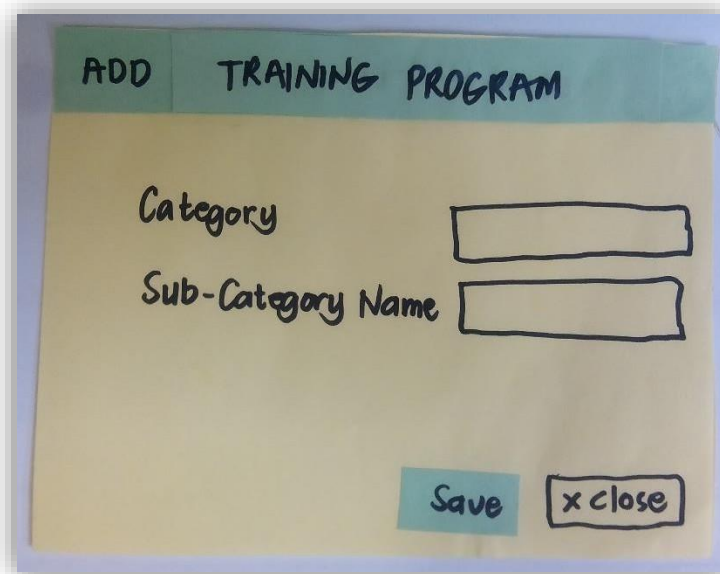


Figure 17. Training Program Registration Prototype

3.2.17 Training Notifications Use case

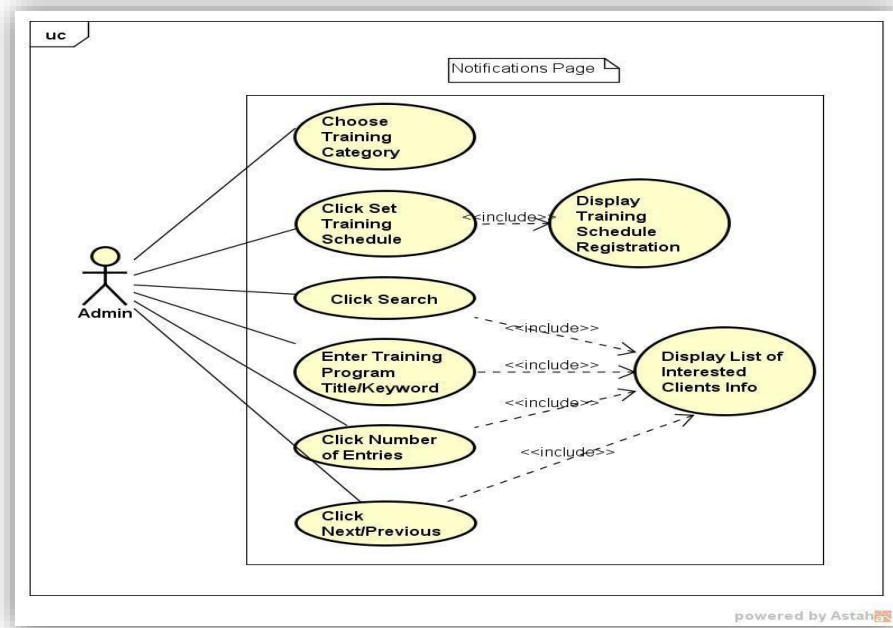


Figure 18. Training Notifications Use Case

3.2.18 Training Notifications Prototype

The prototype shows a 'Train Me' header with a navigation bar containing 'Clients', 'Trainers', 'Training Needs', 'Notifications', 'Reports', 'Settings', 'Profile', and 'Logout'. The 'Notifications' section is active, displaying a dropdown menu for 'Agriculture' and a 'Search' button. Below the search bar, there is a '+ SET TRAINING SCHEDULE' button. The main content area shows 'Show 5 records' and a table of training notifications. The table has columns for 'TRAINING CATEGORY', 'TRAINING SUB-CATEGORY', 'CLIENT NAME', 'ADDRESS', 'Category', and 'CONTACT NUMBER'. The search results are filtered by 'Organic Farming'.

TRAINING CATEGORY	TRAINING SUB-CATEGORY	CLIENT NAME	ADDRESS	Category	CONTACT NUMBER
Agriculture	Organic Farming	Roger Aguilar	Abaca, LE	Farmer	0915 111111
Agriculture	Organic Farming	Edmund Baco	Garita, SE	Farmer	0916 111110
Agriculture	Organic Farming	Francis Hinaya	Baculan, S.E	OSY	0917 111110
Agriculture	Organic Farming	Art Defensor	Pob-Ilaya, SE	Farmer	0918 111101
Agriculture	Organic Farming	Oscar Garin	Camiri, SE	OSY	0919 111100

Showing 5 of 215 records

Previous 1 2 Next

Figure 19. Borrowed Page Prototype

3.2.19 Set Training Schedule Use case

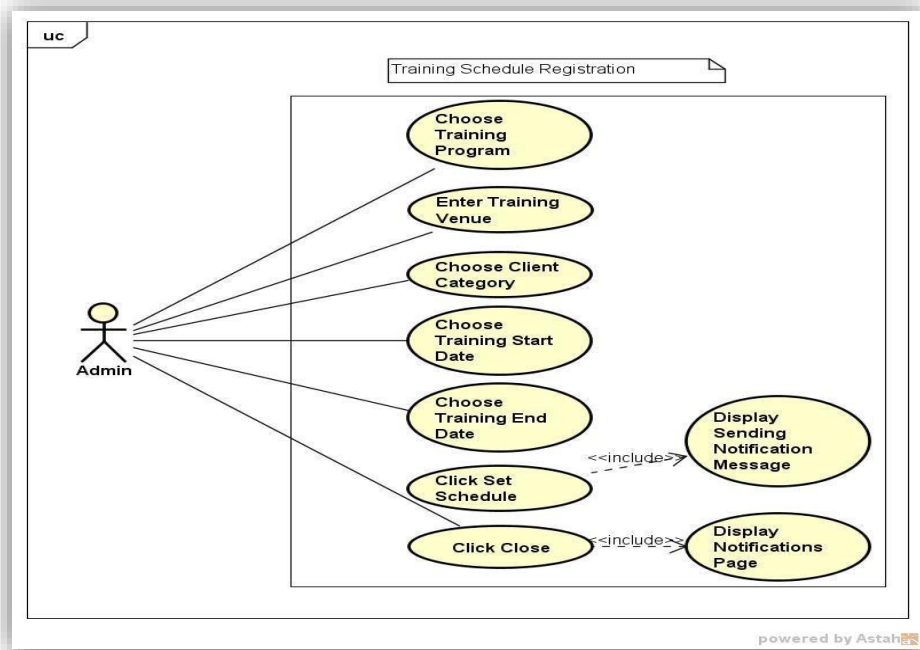


Figure 20. Set Training Schedule Use Case

3.2.20 Set Training Schedule Prototype

A hand-drawn prototype of the "SET TRAINING SCHEDULE" form. The form is titled "SET TRAINING SCHEDULE" and contains the following fields:

- Client Category
- Training Category
- Training Sub-Category
- Training Venue
- Training START Date
- Training END Date

At the bottom of the form, there are two buttons: "Set Schedule" and "x close".

Figure 21. Set Training Schedule Prototype

3.2.21 Reports Page Use case

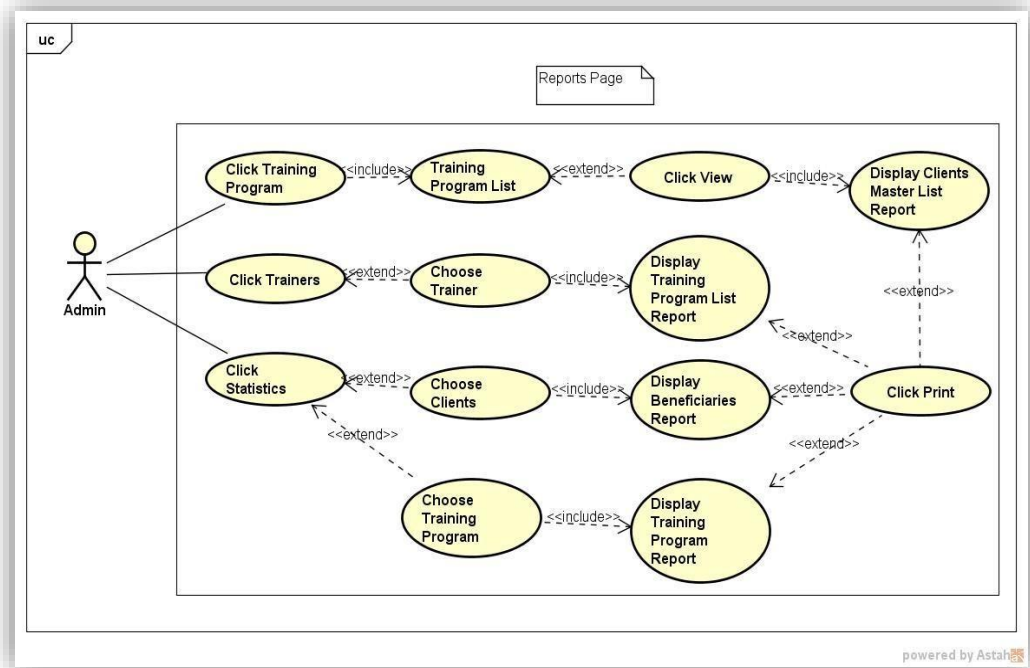


Figure 22. Reports Page Use case

3.2.22 Reports Page Prototype

Hand-drawn prototype of the Reports Page:

Train Me

Navigation: Clients | Trainers | Training Needs | Notifications | Reports | Settings | Profile | Logout

REPORTS

TRAINING PROGRAM | TRAINERS | STATISTICS

TRAINING CAT.	TRAINING PROGRAM	VENUE	START DATE	END DATE	ACTION
Agriculture	Organic Farming	Cubay Brgy. Hall	8/10/17	8/20/17	View
IT	Personnel Profile System	ISCOF-SEC Clab	10/20/17	10/30/17	View
Livelihood	Trash to Cash	Garita Brgy. Hall	9/5/17	9/15/17	View
Education	Functional Literacy	Dacal Brgy. Hall	10/20/17	10/30/17	View
TESDA	Defensive Driving	ISCOF-SEC	10/20/17	10/30/17	View

Figure 23. Reports Page Prototype

3.2.23 Settings Page Use case

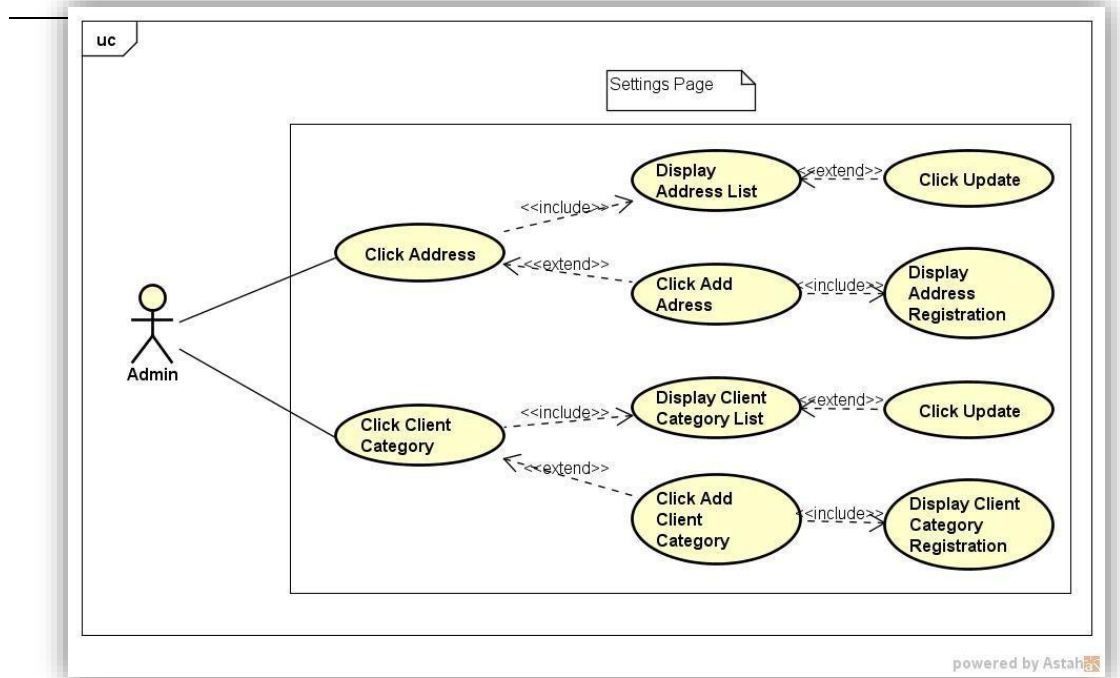


Figure 24. Settings Page Use case

3.2.24 Settings Page Prototype

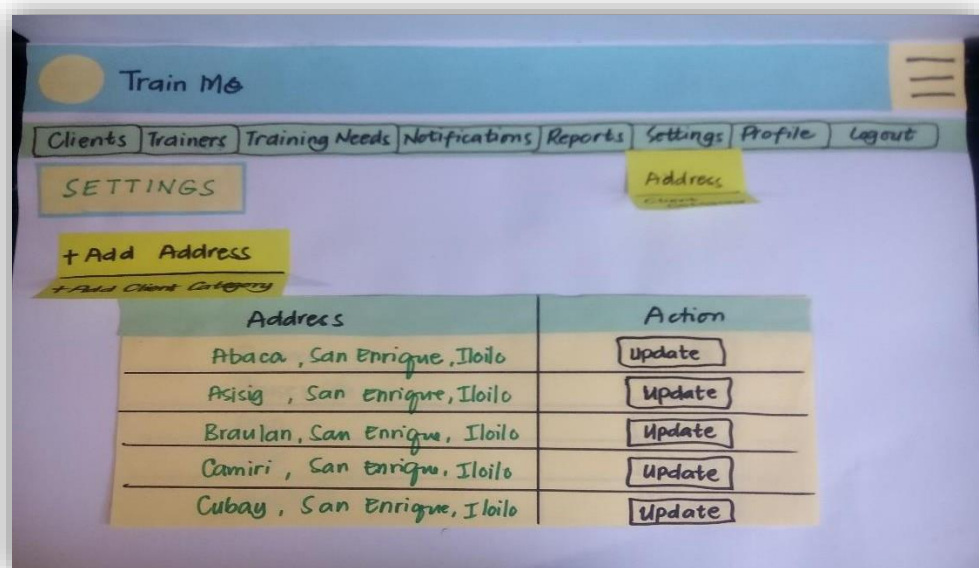


Figure 25. Settings Page Prototype

3.2.25 Client Login Use case

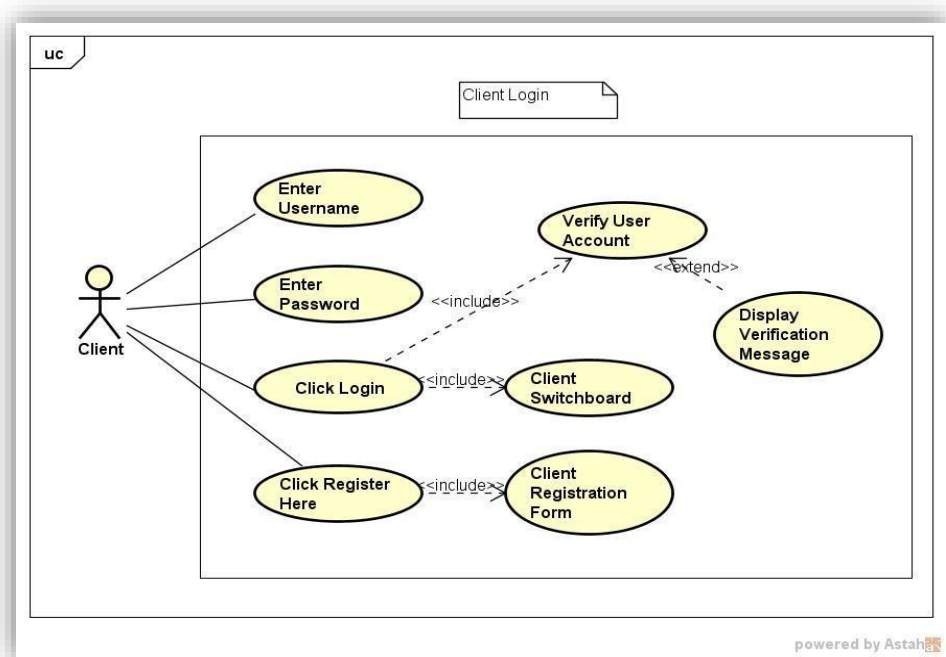


Figure 26. Client Login Use case

3.2.26 Client Login Prototype



Figure 27. Client Login Prototype

3.2.27 Client Login Use case

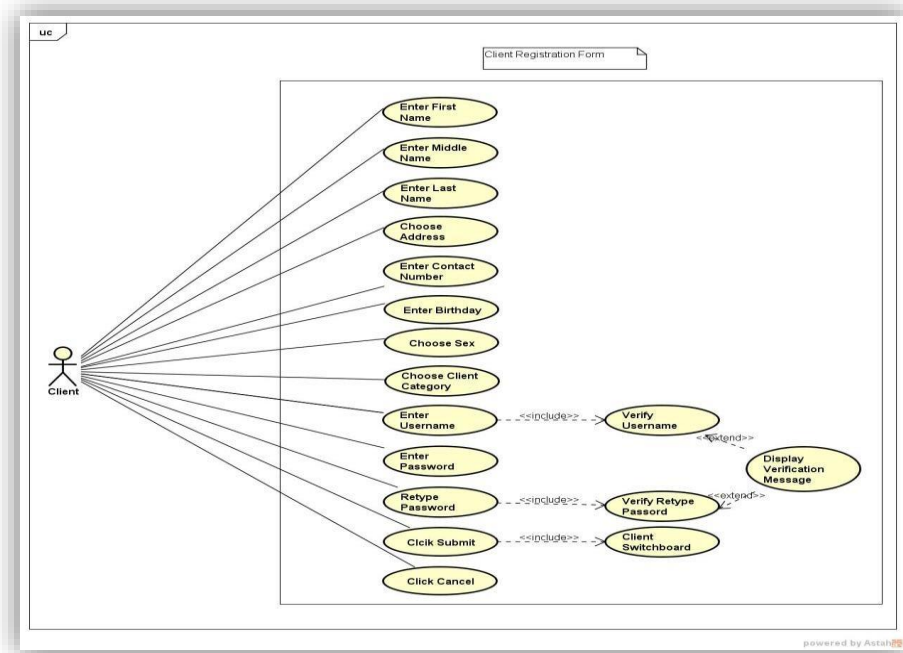


Figure 28. Client Registration Use case

3.2.28 Client Registration Prototype

A hand-drawn prototype of a client registration form titled "REGISTRATION Form". The form contains the following fields and values:

ID Number	1
First Name	Roger
Middle Name	Cajandig
Last Name	Aguilar
Address	Abaca, San Enrique, Iloilo ▼
Contact Number	09154678321
Birthday	2/14/1970
Sex	M ▼
Client Category	Farmer ▼
Username	groovyfarmer
Password	*****
Retype Password	*****

At the bottom of the form are two buttons: "Submit" and "Cancel".

Figure 29. Client Registration Prototype

3.2.29 Client Home Page Use case

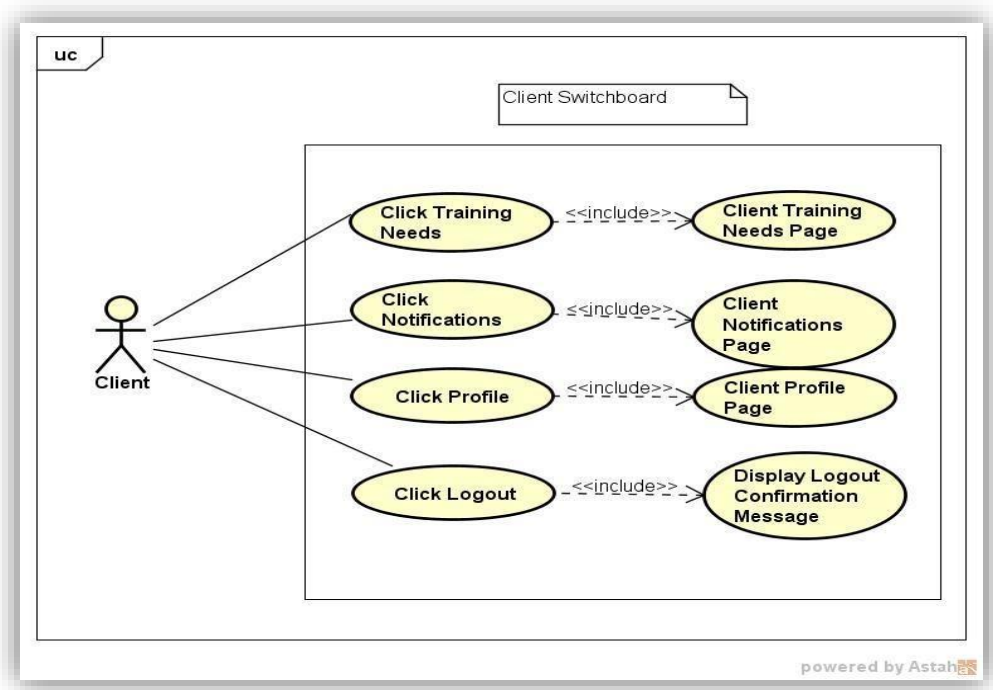


Figure 30. Client Home Page Use Case

3.2.30 Client Home Page Prototype

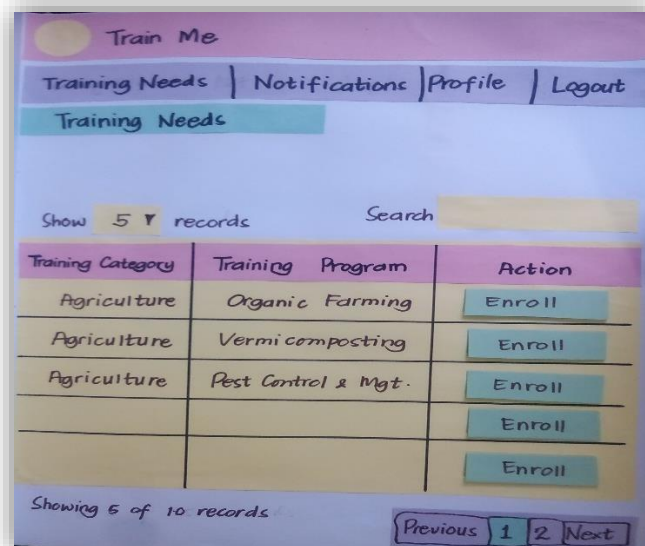


Figure 31. Client Home Page Prototype

3.2.31 Client Training Needs Use case

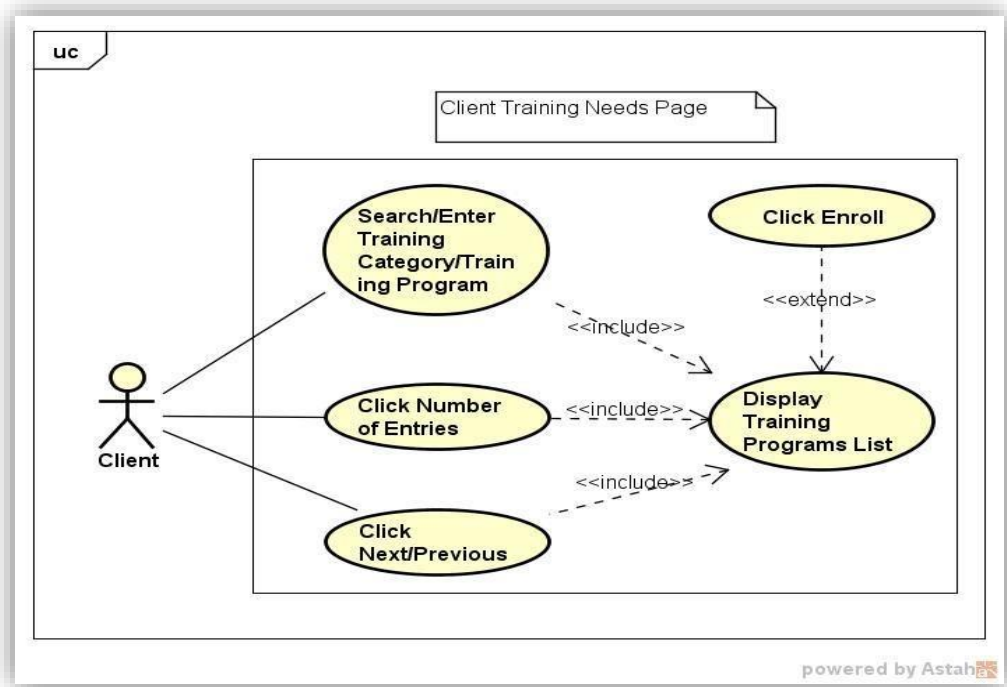


Figure 32. Client Training Needs Use Case

3.2.32 Client Training Needs Prototype

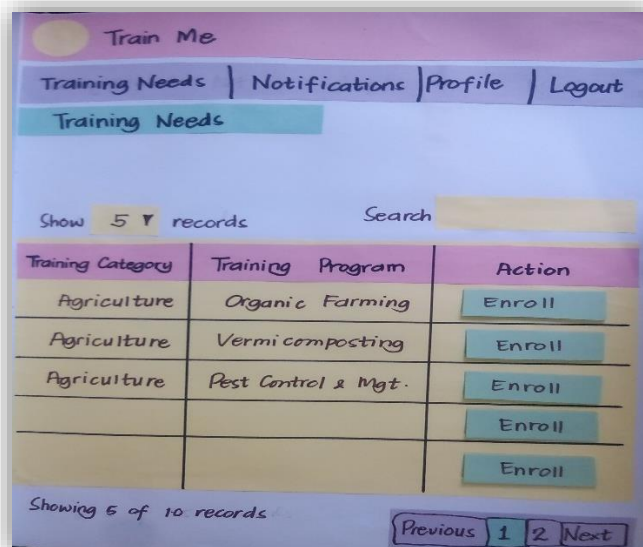


Figure 33. Client Training Needs Prototype

3.2.33 Client Notifications Use case

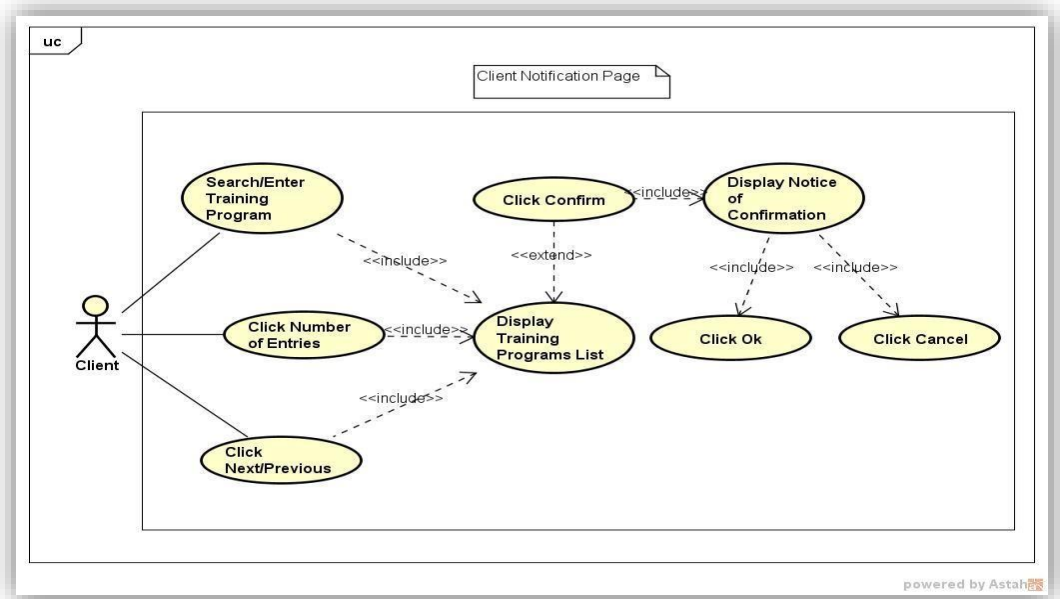


Figure 34. Client Notifications Use Case

3.2.34 Client Notifications Prototype

Train Me

Training Needs | Notifications | Profile | Logout

Training NOTIFICATIONS

Show 5 ▼ records Search

Training Program	Venue	Start Date	End Date	Action
Organic Farming	Cubay Brgy Hall	8/10/2017	8/20/2017	Confirm
				Confirm
				Confirm
				Confirm
				Confirm

Showing 5 of 7 records

Previous 1 2 Next

Figure 35. Client Notifications Prototype

3.2.35 Trainer Login Use case

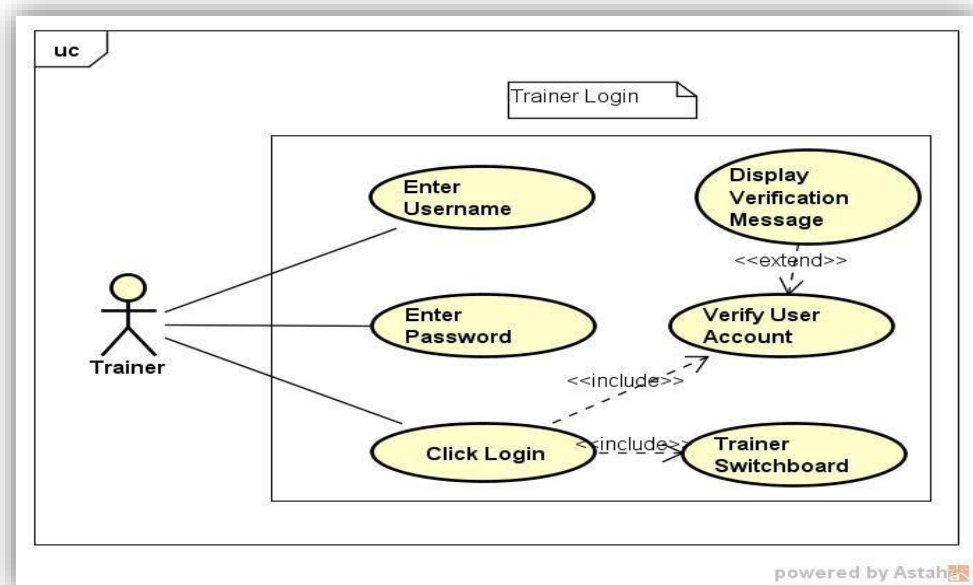


Figure 36. Trainer Login Use case

3.2.36 Client Notifications Prototype

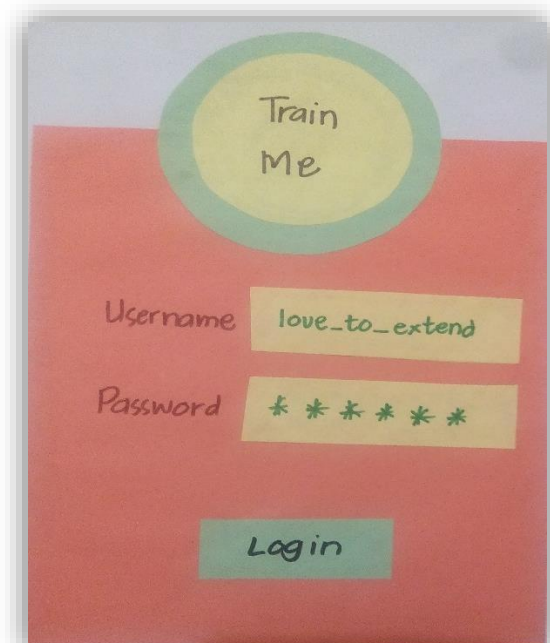


Figure 37. Trainer Login Prototype

3.2.37 Trainer Home Page Use case

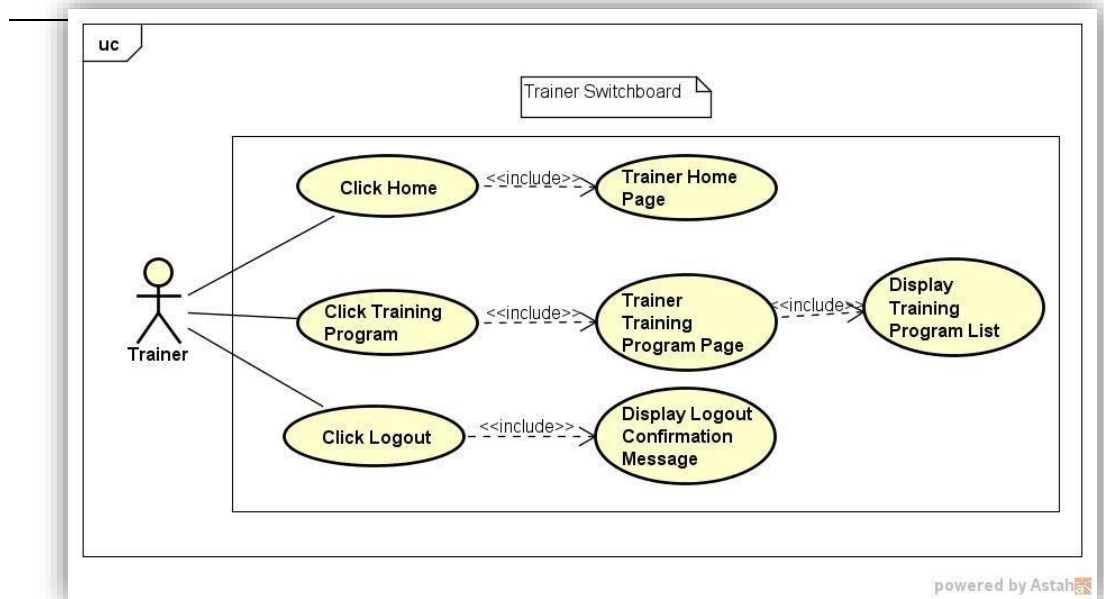


Figure 38. Trainer Home Page Use case

3.2.38 Trainer Home Page Prototype



Figure 39. Trainer Home Page Prototype

3.2.39 Trainer Training Program Use case

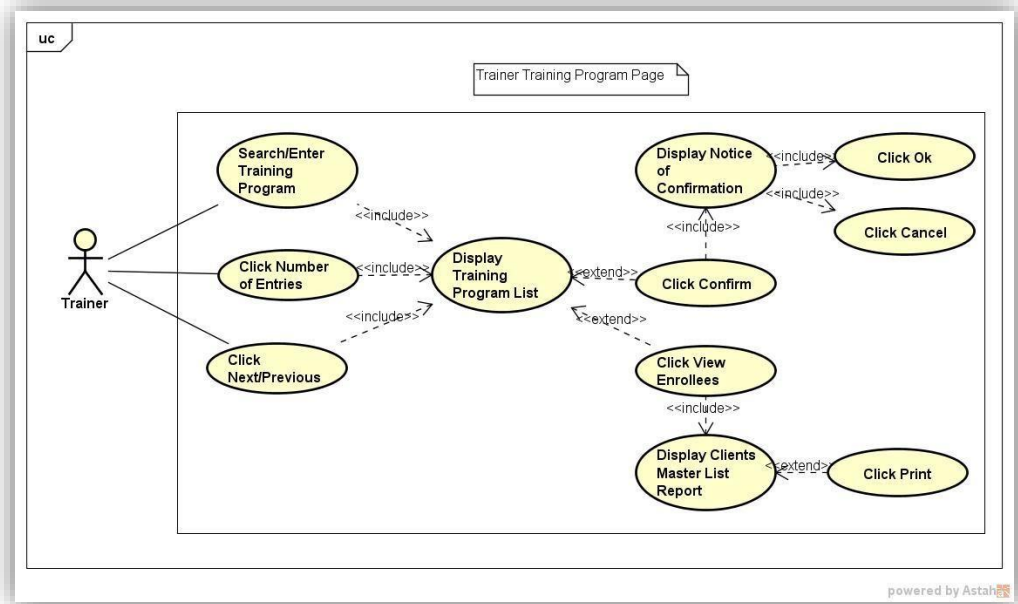


Figure 40. Trainer Training Program Use case

3.2.40 Trainer Home Page Prototype

CLIENTS Master List

Training Program: Organic Farming
 Venue: Cubay Brgy. Hall
 Start Date: 10/10/2017 End Date: 10/20/2017
 Client Category: Farmers
 Trainers: K. Padilla, M. Prudente, N. Gonturo

No.	Client Name	Contact Number
1	Roger C. Aguilar	0915 4678901
2	Erick L. Lapating	0916 5678921
3	Sixto M. Payas	0917 2935560
4	Dexter M. Yuralde	0920 1111444
5	Pedro P. Lamera	0918 3432367
6	Roy N. Solivio	0916 2968012
:		

PRINT Report

NOTICE of Confirmation

This is to confirm ~~up~~ that you will be a TRAINER for Trash to Cash Training with the PWDs on October 10, 2017 to October 20, 2017, 8:00 AM at Barangay Hall, Cubay, San Enrique.

Yes ☐ xCancel ☐

Figure 41. Trainer Home Page Prototype

3.3Performance Requirements

TrainME is likely to store a large amount of clients, trainers, and training programs information that to maximize user experience, the system should perform as efficiently as expected. Generally, TrainME should be capable of:

- Processing transactions in a fast and accurate manner.
- Accommodating large amounts of data and information.
- Handling invalid combination of usernames and passwords.

Specifically, TrainME is divided into three user accounts, each of which should perform as outlined below:

3.3.1 Admin Account

3.3.1.1 Add/ Update Trainers, Training Categories, Training Programs

Data DESC: This feature should be evident as this is the springboard of every transaction. RAT: In order to guide the admin as to what data should be added or updated.

3.3.1.2 Set Training Schedules and Send Notifications

DESC: This feature should be prominent as this determines the actual clients to attend the training program.

RAT: In order to guide the admin on what training program to launch and what client category should receive notifications.

3.3.1.4 Generate Reports

DESC: This feature should be highlighted as it categorizes, summarizes and organizes the clients, trainers and training programs data into meaningful information.

RAT: In order to guide the admin as to what outputs are available for hardcopy.

3.3.2 Client Account

3.3.2.1 Register and Update Profile

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DESC: This feature should be prominent as this is the launching point of client transaction. RAT: In order to give the clients a convenient way of making them known to the trainers.

3.3.2.2 Submit Training Need

DESC: This feature should be prominent as this is the heart of client's transaction. RAT: In order to give the clients an efficient way of letting their training needs be heard.

3.3.2.1 Receive Notifications

DESC: This feature should be visible as this the official and valuable notice for clients as far as participation in training programs is concerned.

RAT: In order to provide the clients with some form of response to their expression of interest in attending a particular training program.

3.3.3 Trainer Account

3.3.3.1 Change Default Username and Password

DESC: This feature should be noticeable as the trainers may opt to use their desired username and password.

RAT: In order to give the trainers a sense of unique identity in utilizing the system.

3.3.3.2 Receive Notifications

DESC: This feature should be prominent as this serves as the springboard of information interchange between the admin and trainers.

RAT: In order to inform the trainers the training programs to be conducted that requires their expertise.

3.3.3.3 Generate Reports

DESC: This feature should be highlighted as it categorizes, summarizes and organizes the clients and training programs data into meaningful information.

RAT: In order to guide the trainers an overview of the list of their clients and as well as a printable document that may be utilized as exhibits.

3.4 Design Constraints

TrainME follows standard design rules and is primarily governed by the operating system and hardware specifications installed.

3.5 Software System Attributes

TrainME provides users with assurance that their account is protected by providing them with:

- *Account authentication and validation using a unique user/member ID.*
- *Monitoring system to update account status pop up reminders for book return overdue dates and fines.*
- *Users' accountability which includes not divulging the member's account to another member.*

3.6 Other Requirements

3.6.1 Safety Requirements

Virus is a harmful computer program that may infect the TrainME database or the operating system as a whole. Thus, a regular system backup is highly encouraged. Crash errors may also be prevented by providing an alternative power supply facility.

3.6.2 Security Requirements

Clients data as well as training information and attachments are valuable that they should be protected at all times. Likewise, it takes a considerable amount of resources to develop the system as whole that its security should be prioritized. Security requirements are listed as follows:

- The database is password-protected.
- Clients and trainers can only read information but cannot edit or modify the system.
- TrainME will have different types of users and each user has access constraints.
- Proper user authentication is provided.
- Other users will not be able to hack user password.
- Admin and member users have different user interface, thus

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only the admin can access the database.